



PROFESSIONAL CERTIFICATION

Professional C++ Programming Training

Elevate your coding skills with Professional C++ Programming. This course covers modern C++ standards advanced memory management templates multithreading and design patterns. Build high-performance robust applications used in real-world industries like gaming finance and systems programming.



30

Hours



Certified

Trainers



Industry

Expert



Professional Training | Industry-Recognized Certificate

Dubai, UAE

About Orbit Training Center

Your Gateway to Professional Excellence in Dubai



Who We Are

Established in 2023, Orbit Training Center is a leading professional training provider in Dubai, offering 60+ professional certification courses across diverse fields, including engineering, IT, design, and languages.



Our Benefits

-  Industry recognized certifications
-  Convenient location in the heart of Dubai
-  Affordable fees with multiple payment plans
-  Lifetime post-training support

★Our Features

-  Hands-on learning with real-world projects
-  Expert trainers with industry experience
-  Flexible learning options to fit your schedule
-  State-of-the-art infrastructure & facilities



Our Mission

We deliver outstanding courses, expert teaching, strong career guidance, and placement support to help every student achieve their best results. We stay connected with alumni through networking events, lifelong learning, and ongoing mentorship. We run our operations with smart systems and data-driven decisions to create a smooth, exceptional experience for all.

Professional C++ Programming

Master professional C++ techniques for building efficient and scalable software applications.

Course Description

This Professional C++ Programming course takes you from intermediate to expert level. Learn modern C++17/20 features STL optimization performance tuning and secure coding practices. Develop production-ready applications through hands-on projects that prepare you for top-tier software engineering roles.

Duration

30

Hours of Training

Format

Classroom / Online

Flexible Option

Key Outcomes

- ✓ Write high-performance C++ code
- ✓ Master advanced memory management
- ✓ Implement modern C++ features
- ✓ Design scalable applications
- ✓ Apply multithreading effectively
- ✓ Use STL and design patterns
- ✓ Build complete professional projects

Target Audience

- Software developers advancing C++ skills
- Game and systems programming engineers
- Computer science graduates entering industry

Certification

- Certificate of Course Completion
- Industry Recognized Certification
- Trained by Certified Professionals

Prerequisites

- Solid understanding of basic C++
- Knowledge of object-oriented programming
- Access to modern C++ compiler

Professional C++ Programming

Start your Professional C++ journey with in-depth modules covering core concepts to advanced real-world application development.

01 Introduction to C++ and Development Setup

- > What C++ is and where it is used games systems embedded
- > Installing a C++ compiler GCC MinGW and setting up VS Code
- > Writing compiling and running your first C++ program
- > Understanding the basic structure of a C++ program
- > How C++ differs from other languages like Python and JavaScript

02 Variables Data Types and Input Output

- > Declaring variables with int float double char and bool
- > Understanding variable naming rules and best practices
- > Using cin and cout for reading input and printing output
- > Constants with const and compile-time values with #define
- > Type conversion (implicit and explicit casting in C++)

03 Operators and Expressions

- > Arithmetic operators (add subtract multiply divide modulus)
- > Relational operators (equal not equal greater less than)
- > Logical operators (AND OR and NOT for conditions)
- > Increment and decrement operators prefix and postfix
- > Assignment operators and compound assignment shortcuts

04 Control Flow (Conditionals and Switch)

- > if else if and else for decision-making in programs
- > Nested if statements for multiple condition checks
- > switch case for matching specific values efficiently
- > Ternary operator for short one-line conditional expressions
- > Common mistakes in conditionals and how to avoid them

05 Loops (Repetition and Iteration)

- > for loop (syntax initialization condition and increment)
- > while loop for repeating while a condition is true
- > do-while loop for executing at least once before checking
- > Nested loops for working with patterns and grids
- > break and continue to control loop flow and exit

06 Functions in C++

- > Defining and calling functions with return types and parameters
- > Pass by value (how copies are passed into functions)
- > Pass by reference (modifying original variables in functions)
- > Function overloading (same name different parameters)
- > Default parameter values and inline functions in C++

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07 Arrays and Strings

- > Declaring and accessing single-dimensional arrays in C++
- > Multi-dimensional arrays and working with 2D grids
- > Common array operations (traverse search and sort)
- > C-style strings and character arrays with null terminator
- > Using the `std::string` class and string functions in C++

08 Pointers and Memory Basics

- > What pointers are and how they store memory addresses
- > Declaring dereferencing and using pointers correctly
- > Relationship between pointers and arrays in memory
- > Dynamic memory allocation with `new` and `delete` operators
- > Common pointer errors (null dangling and memory leaks)

09 Object-Oriented Programming (Classes and Objects)

- > What OOP is and why it is used in C++ programs
- > Defining a class with data members and member functions
- > Creating objects and accessing members with dot operator
- > Access specifiers (public private and protected explained)
- > Constructors for initializing objects when they are created

10 OOP (Encapsulation and Getters Setters)

- > Encapsulation (hiding data and exposing only what is needed)
- > Writing getter and setter methods for private members
- > Destructors (cleaning up resources when object is destroyed)
- > The `this` pointer (referring to the current object)
- > Separating class declaration into header and source files

11 OOP (Inheritance)

- > What inheritance is and how it promotes code reuse
- > Defining a derived class that inherits from a base class
- > Overriding base class functions in the derived class
- > Types of inheritance (single multilevel and hierarchical)
- > Using protected members to share data with derived classes

12 OOP (Polymorphism and Virtual Functions)

- > What polymorphism is and how it works in C++
- > Function overriding and calling the correct version at runtime
- > Virtual functions and the `virtual` keyword explained
- > Pure virtual functions and abstract base classes
- > Using base class pointers to call derived class methods

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13 Standard Template Library (STL Basics)

- > What the STL is (containers iterators and algorithms)
- > Using vector as a dynamic resizable array in C++
- > Using map for key-value pair storage and lookup
- > Using set for storing unique values automatically
- > Common STL algorithms (sort find count and reverse)

15 Exception Handling and Error Management

- > What exceptions are and why error handling matters
- > Using try catch and throw for handling runtime errors
- > Catching different exception types with multiple catch blocks
- > Standard exception classes from the stdexcept header
- > Writing safe and robust code using exception best practices

14 File Handling in C++

- > Opening reading and closing text files with ifstream
- > Writing and appending data to files with ofstream
- > Reading files line by line using getline function
- > Checking file open success and handling file errors
- > Reading and writing structured data to text files

16 Final Project and C++ Best Practices

- > Planning and building a complete C++ console application
- > Applying OOP principles — classes inheritance and polymorphism
- > Using STL containers and algorithms in the final project
- > Code organization (header files source files and namespaces)
- > Debugging tips and writing clean readable C++ code

Final Projects and Assessment:

- **Assessment:** Project-based evaluation through real-world projects to assess practical skills and application without theory exams.
- **Tailored Curriculum:** Customized to industry standards, covering essential topics for professional development in relevant fields.
- **Hands-on Projects:** Real-world simulations and interactive tasks to build applicable skills and create professional portfolios.
- **Certification Prep:** Targeted review sessions, practice exercises, and strategies for achieving industry certifications.
- **Ongoing Support:** Access to recorded materials, forums, instructor guidance, and career assistance for long-term growth.

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Muppuli Raja



One of great institute for REVIT BIM course. I completed my training with Mr Ahmed. He is one great trainer in this institute. He has very good knowledge for BIM Training. I completed my REVIT BIM training with orbit and it was very nice learning journey for me. And one of best thing is training center is very flexible for classes.



Arun



Best SolidWorks training in Dubai! The instructor made 3D modeling and assembly design easy to understand. The course covered all essential tools I need for product design and engineering work. Best training center for solidworks training. Best Training Center for Best Skills



Harshu CH



My experience at Orbit training center has been the best In every step I learn from there very clearly almost complete my course it's very interesting the trainer is the best professional and friendly ,and the rest of the team keep it up, thanks for supporting me Orbit and all staff...



Ranjan Axalan



I just finished my course in 3ds max. I learned a lot on how to create 3d model. Thank you for the experience and for the teaching method it help me to understand easily and grow my knowledge. I will use my learnings to my future company. Highly Recomanded!



Numan



I had good support from trainer Mr Ahmed and the other staff. They would schedule my BIM classes as per my convenience and would retake classes if I required. They always answer my calls when I need assistance. Not only that the trainer would go a mile a ahead by helping me with my work related issues like a true mentor.



Lelanie Fernandez



Overall, I highly recommend the Fusion 360 training at Orbit Center, especially with Ahmad as the instructor. This course has exceeded my expectations, and I am learning a lot. I would rate it 5 out of 5 stars!

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Aman Bakhtiyar



I had a great learning experience at Orbit Training Center. The trainers are knowledgeable, supportive, and always ready to help. The clear and practical teaching methods really helped improve my skills and confidence. I highly recommend Orbit Training Center to anyone looking to learn AutoCAD, Revit, and other professional skills.



Sana Zahid -



I recently completed the online Digital Marketing course with Orbit Training Center, and it was a great experience! The lessons were clear, practical, and easy to follow. The trainers were knowledgeable and supportive throughout the course. I've learned a lot that I can apply to my work here in Belgium. Highly recommend Orbit Training Center to anyone looking to improve their digital marketing skills!



Cleavon Henderson



Orbit Training Center offers outstanding CAD classes that are easy to understand, even as a first-year mechanical engineering student. The instructors simplify complex concepts and use practical examples, making the learning process engaging and effective. Whether you're new to CAD or improving your skills, Orbit Training Center provides a supportive and enriching environment. Highly recommended!



Danish Khan



Just finished Primavera P6 training at Orbit Training Center and it was fantastic! The trainer made complex scheduling concepts easy to understand with real-world examples. Got hands-on experience with project planning, resource allocation, and progress tracking. The course was well-structured and I feel confident using Primavera P6 for my projects now. Great value and would definitely recommend to other project managers!



500+ Students Trained

Join our growing community of professionals

95%

Satisfaction Rate

60+

Courses Offered

4.6

Average Rating



GET STARTED TODAY

Get in Touch with Us

At Orbit Training Center, we offer a diverse range of professional courses designed to equip you with the latest industry skills and knowledge. Our comprehensive programs cover in-demand fields such as Interior Design, Computer Science, Graphic Design, Digital Marketing, Solar Energy, and specialized software training including STAAD Pro, Revit Architecture, Tekla Structures, and more. Each course is crafted by industry experts to provide hands-on learning, real-world applications, and certification preparation.

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Free

Demo Session



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